

RING, FIELD AND VECTOR SPACES

Ring:

A set R with two binary operations addition '+' and multiplication '×' is called a ring if

- (i) $(R, +)$ is abelian group.
- (ii) $(R, \times)$ is a semi group.
- (iii) R satisfies the distributives laws.

The following table shows the information of some standard Rings.

Rings
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
$M_n = \{\text{set of all } n \times n \text{ matrices with real entries}\}$
$\mathbb{Z}_n = \{\text{Set of all residue classes of } n\}$
$2\mathbb{Z} = \{\text{Set of all even integers}\}$
$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ Ring of Gaussian integers
$P[x] = \{\text{Set of all polynomials in } x \text{ with coefficients in } P\}$ Where P can be $\mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$.

Subring:

A subring of a ring R is the subset of R which is ring itself under the same binary operations of addition and multiplication. For example $\mathbb{Q}$ is a subring of $\mathbb{R}$.

Ideal:

Let R be a commutative ring and $I \subset R$ be its subring. Then $I \subset R$ is called an ideal if for any $r \in R$ and $x \in I$ we have $rx \in I$. For example $2\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Integral Domain:

A ring R is called integral domain if

- (i) $(R, \times)$ is commutative.
- (ii) R has multiplicative identity.
- (iii) R is without zero divisor.

The following table shows the information of some integral domains.

Integral Domains
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
$\mathbb{Z}_p = \{\text{Set of all residue classes of } p\} \text{ where } p \text{ is a prime number}$
$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ Ring of Gaussian integers
$\mathbb{Q}[x] = \{\text{Set of all polynomials in } x \text{ with coefficients in } \mathbb{Q}\}$

Division Ring:

A ring R is called division ring if $(R, \times)$ is a group under multiplication. For example $M_n = \{\text{set of all } n \times n \text{ matrices with real entries}\}$ is a division ring.

Unique Factorization Domain:

An integral domain R is called a unique factorization domain (UFD) if

- (i) Every non-zero element of R is either unit or a product of finitely many irreducibles.
- (ii) If $a_1, a_2, \dots, a_m = b_1, b_2, \dots, b_n$, where the a_i, b_j are irreducibles, then $n = m$ and after reordering of factors, a_i and b_i are associates for $1 \leq i \leq n$.

For example $\mathbb{Z}$ is a unique factorization domain.

Euclidean Domain:

An integral domain R is called a Euclidean domain if there exists Euclidean domain function $\varphi : R \setminus \{0\} \rightarrow \mathbb{Z} \geq 0$ satisfying the following conditions:

- (i) for all non-zero $a, b \in R$, we have $\varphi(a) \leq \varphi(ab)$,
- (ii) given $a, b \in R$, $b \neq 0$, there exist $q, r \in R$ such that $a = qb + r$ and $r = 0$ or $\varphi(r) < \varphi(b)$.

For example $\mathbb{Z}$ with $\varphi(x) = |x|$ is a Euclidean domain.

Principal Ideal Domain:

An ideal of R of the form aR (the multiples of a given element $a \in R$) is called a principal ideal. An integral domain R is called a principal ideal domain if every ideal of R is principal. For example $\mathbb{Z}$ is a principal ideal domain.

Field:

A set F with two binary operations addition '+' and multiplication '×' is called a field if

- (iv) $(F, +)$ is abelian group.
- (v) $(F - \{0\}, \times)$ is an abelian group.
- (vi) F satisfies the distributives laws.

The following table shows the information of some standard fields.

Fields
$\{0, 1\}$
$\mathbb{Q}, \mathbb{R}, \mathbb{C}$
$\mathbb{Z}_p = \{\text{Set of all residue classes of } p\}$ where p is a prime number.
Power set $P(X)$ with union and intersection

Some Important Theorems

1. A finite integral domain is a field.
2. Every field is an integral domain.
3. A commutative division ring is a field.
4. Every Euclidean domain is a principal ideal domain.
5. Every principal ideal domain is a unique factorization domain.

Vector Space:

A vector space is a set V on which two operations $+$ and $\cdot$ are defined, called vector addition and scalar multiplication.

The operation $+$ (vector addition) must satisfy the following conditions:

- (i) Closure: If u and v are any vectors in V , then the sum $u + v$ belongs to V .
- (ii) Commutative law: For all vectors u and v in V , $u + v = v + u$
- (iii) Associative law: For all vectors u, v, w in V , $u + (v + w) = (u + v) + w$
- (iv) Additive identity: The set V contains an additive identity element, denoted by 0 , such that for any vector v in V , $0 + v = v$ and $v + 0 = v$.
- (v) Additive inverses: For each vector v in V , the equations $v + x = 0$ and $x + v = 0$ have a solution x in V , called an additive inverse of v , and denoted by $-v$.

The operation $\cdot$ (scalar multiplication) is defined between real numbers (or scalars) and vectors, and must satisfy the following conditions:

- (vi) Closure: If v is any vector in V , and c is any real number, then the product $c \cdot v$ belongs to V .
- (vii) Distributive law: For all real numbers c and all vectors u, v in V , $c \cdot (u + v) = c \cdot u + c \cdot v$
- (viii) Distributive law: For all real numbers c, d and all vectors v in V , $(c+d) \cdot v = c \cdot v + d \cdot v$
- (ix) Associative law: For all real numbers c, d and all vectors v in V , $c \cdot (d \cdot v) = (cd) \cdot v$
- (x) Unitary law: For all vectors v in V , $1 \cdot v = v$

Subspace:

Let V be a vector space, and let W be a subset of V . If W is a vector space with respect to the operations in V , then W is called a subspace of V .

The following table shows the information of some standard vector spaces with their underlying fields.

Vector Space	Field
Q	Q
R	Q, R
C	Q, R, C
$M_n = \{\text{Set of all } n \times n \text{ non-singular matrix matrices}\}$	Q, R, C
$Z_p = \{\text{Set of all residue classes of } p \text{ where } p \text{ is a prime number}\}$	Z_p
$P(X) = \{\text{Set of all subsets of } X\}$	Q, R, C
$L_2 = \{\text{space of all square integrable functions}\}$	R

Multiple Choice Questions

1. Which of the following is a ring.
(a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) all of these
2. Which of the following is a smallest ring.
(a) $\{1\}$ (b) $\{0\}$ (c) $\{0, 1\}$ (d) $\mathbb{Z}$
3. Let R be a ring and $x, y \in R$ such $x \neq 0, y \neq 0$ but $(x)(y) = 0$ then x and y are called:
(a) units (b) scalars (c) zero divisors (d) none
4. Which of the following is a zero divisor in a residue class of 6 i.e., $\mathbb{Z}_6 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$:
(a) 1 (b) 2 (c) 4 (d) 5
5. A ring without zero divisors is called:
(a) commutative ring (b) division ring (c) integral domain (d) field
6. Which of the following is an integral domain.
(a) $\mathbb{Z}_4$ (b) $\mathbb{Z}_6$ (c) $\mathbb{Z}_7$ (d) $\mathbb{Z}_8$
7. Finite integral domain is:
(a) division ring (b) skew field (c) field (d) none
8. If a ring R with unity satisfies the multiplicative property of inverse then R is called:
(a) integral domain (b) division ring (c) field (d) semi-ring
9. A commutative division ring is called:
(a) integral domain (b) skew field (c) field (d) none
10. Which of the following is a division ring:
(a) $\mathbb{Z}_5 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ (b) $\mathbb{Q}$
(c) $M = \{A/A \text{ is an invertible matrix}\}$ (d) $\mathbb{C}$
11. Which of the following is not a field.
(a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) $\mathbb{C}$
12. Which of the following is a field.
(a) $\mathbb{Z}_5$ (b) $\mathbb{Z}_9$ (c) $\mathbb{Z}_3$ (d) all of these
13. The set of residue classes $\mathbb{Z}_p$ is a field if p is:
(a) even (b) odd (c) prime (d) integer
14. The power set $P(X)$ is a field with addition as union and multiplication as:
(a) simple multiplication (b) matrix multiplication
(c) vector multiplication (d) intersection
15. Let S be a nonempty subset of a ring R such that $\forall a, b \in S, a - b, ab \in S$ then S is called:
(a) field (b) subfield (c) subring (d) ideal
16. $\mathbb{Z}$ is a subring of:
(a) $\mathbb{Q}$ (b) $\mathbb{R}$ (c) $\mathbb{C}$ (d) all of these
17. Let S be a non-empty subset of a ring R such that $\forall a \in R$ and $b \in S, ab \in S$ then S is called:

- (a) subfield (b) ideal (c) subring (d) none
18. If I and J are ideal of R then which of the following is also an ideal of R .
 (a) $I + J$ (b) IJ (c) $I \cap J$ (d) all of these
19. Let X be a ring such that all the subring of X are also ideal of X . Then X is:
 (a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{C}$ (d) $\mathbb{R}$
20. A field without a proper subfield is called:
 (a) prime field (b) simple field (c) quotient field (d) none
21. Which of the following is a prime field.
 (a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) $\mathbb{C}$
22. Let X be a vector space over itself then X is:
 (a) group (b) ring (c) integral domain (d) field
23. Which of the following is a vector space over $\mathbb{Q}$
 (a) $\mathbb{Q}$ (b) $\mathbb{R}$ (c) $\mathbb{C}$ (d) all of these
24. Which of the following is a subspace of $\mathbb{R}$ over $\mathbb{Q}$:
 (a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{C}$ (d) none
25. Let V be a vector space. Then which of the following is a subspace of V .
 (a) $\{0\}$ (b) $\{1\}$ (c) $\{0, 1\}$ (d) none
26. Let U and W be two subspaces of a vector space V , then which of the following is also a subspace of V .
 (a) $\{0\}$ (b) $U \cap W$ (c) $U + W$ (d) all of these
27. Which of the following set of vector generators $\mathbb{R}^3$
 (a) $\{(1, 2, 3), (0, 1, 2), (0, 0, 1)\}$ (b) $\{(1, 1, 1), (0, 1, 1), (0, 1, -1)\}$
 (c) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ (d) all of these
28. The dimension of a vector space is number of elements in its:
 (a) subspace (b) base (c) centre (d) field
29. Every finite dimensional vector space has a
 (a) spanning set of vectors (b) set of linearly independent vectors
 (c) base (d) all of these
30. If U and W are finite dimensional subspaces of a vector space V then
 $\dim(U + W) + \dim(U \cap W) =$
 (a) $\dim(U)$ (b) $\dim(V)$
 (c) $\dim(U) + \dim(V)$ (d) $\dim(UW)$
31. If U and W are finite dimensional subspaces of a vector space V such that $U \cap W = \{0\}$ then which of the following is true.
 (a) $\dim(V) = \dim(U) + \dim(W)$ (b) $\dim(U + W) = \dim(W)$
 (c) $\dim(V) + \dim(U) + \dim(W) = 0$ (d) all of these
32. Let U and W be 2 dimensional subspace of $\mathbb{R}^3$. Then which of the following is true.
 (a) $U \cap W = \{0\}$ (b) $U \cap W = V$
 (c) $U \cap W \neq \{0\}$ (d) none
33. Which of the following is a basis for $\mathbb{R}^2$.

- (a) $\{(1, 1), (3, 1)\}$ (b) $\{(1, 2), (3, 1)\}$
 (c) $\{(1, 0), (1, 1)\}$ (d) none
34. Which of the following is a basis for R^3 .
 (a) $\{(1, 2, -1), (0, 3, 1), (1, -5, 3)\}$ (b) $\{(2, 4, -3), \{0, 1, 1\}, (0, 1, -1)\}$
 (c) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ (d) all of these
35. A function between two vector spaces over the same field is called:
 (a) transformation (b) operator
 (c) metric (d) matrix
36. Which of the following is not a linear transformation from R^3 to R^2 .
 (a) $T(x_1, x_2, x_3) = (|x_1|, 0)$ (b) $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$
 (c) $T(x_1, x_2, x_3) = (0, x_3)$ (d) all of these
37. If $T: R^2 \rightarrow R$ is linear such that $T(1, 1) = 3$ and $T(0, 1) = -2$ then $T(x_1, x_2) =$
 (a) $x_1 + x_2$ (b) $5x_1 - x_2$
 (c) $5x_1 - 2x_2$ (d) $3x_1 + 4x_2$
38. Which of the following is a matrix of linear transformation $T: R \rightarrow R^2$ defined by $T(x) = (3x, 5x)$.
 (a) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (b) $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ (c) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ (d) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
39. Which of the following is a linear transformation.
 (a) zero transformation (b) identity transformation
 (c) differential transformation (d) all of these
40. Let $\dim(U) = \dim(V) = 50$ and $T: U \rightarrow V$ be linear transformation such that $\dim(R(T)) = 15$. Then $\dim(N(T)) =$
 (a) 25 (b) 20 (c) 35 (d) 50
41. If $T: U \rightarrow V$ is a one-to-one linear transformation if:
 (a) $R(T) = \{0\}$ (b) $N(T) = \{0\}$
 (c) $\dim(U) \leq \dim(V)$ (d) none
42. Let $T: U \rightarrow V$ be linear transformation, then which of the following is a subspace of V .
 (a) $\{0\}$ (b) $R(T)$ (c) V (d) all of these
43. Which of the following is finite dimensional vector space.
 (a) Q (b) R (c) C (d) none
44. $\{(2, 1), (1, -1)\}$ is a basis of:
 (a) R^2 (b) R^3 (c) R^4 (d) R
45. $\forall a, b \in X, f(a + b) = f(a)f(b)$ and $f(a)f(b) = f(ab)$ then f is a:
 (a) group homomorphism (b) ring homomorphism
 (c) vector homomorphism (d) none

46. Let V be a vector space over field F . Then V is abelian w.r to:
(a) addition (b) multiplication
(c) scalar multiplication (d) all of these
47. Let V be a vector space over field F , then V is also:
(a) abelian group w.r.t multiplication (b) ring
(c) field (d) monoid w.r.t addition
48. Which of the following is not a vector space over R .
(a) Z (b) Q (c) N (d) all of these
49. Which of the following is a subspace of C .
(a) $\{1, w, w^2\}$ (b) $\{1, -1, i, -i\}$
(c) $\{a + \sqrt{3}i/a \in N\}$ (d) none
50. Set of all matrices is a vector space over:
(a) Q (b) R (c) C (d) none

Key To MCQ Of Ring, Field and Vector Spaces

1	a	2	C	3	c	4	c	5	C	6	c
7	c	8	B	9	c	10	c	11	A	12	b
13	c	14	D	15	c	16	d	17	B	18	d
19	a	20	A	21	a	22	d	23	D	24	b
25	a	26	D	27	d	28	b	29	D	30	c
31	a	32	C	33	a	34	d	35	A	36	a
37	c	38	B	39	d	40	c	41	B	42	d
43	d	44	A	45	b	46	a	47	D	48	d
49	d	50	D								